

Dino v0.1.1 — Instruction Decoder + 7-Segment Display Upgrade

Objective

Upgrade Dino v0.1.0 by adding SN74LS138N and two 7-segment display drivers.

Additions

- SN74LS138N 3-to-8 Line Decoder
 - Connected to D0–D2 of the SRAM data bus
 - G1 = HIGH, G2A = G2B = LOW
 - Output lines:
 - Y1 → Register A latch (LOAD_A)
 - Y2 → Register B latch (LOAD_B)
 - Y3 → ALU control (ADD)
 - Y5 → Output latch clock (OUT)
- Two 7-Segment Display Drivers
 - 4-bit BCD inputs from 8-bit ALU output bus
 - D0–D3 → right digit
 - D4–D7 → left digit

Reserved Values and Constraints

- Instruction Opcodes: 0x00 through 0x07
- Safe Output Range: 0x08 to 0xFF
- Effective Addition Range: 0x08 to 0x5B (decimal 8–91)

Completion Criteria

- Autonomous execution
- ALU result latched and displayed on 7-segment digits
- Decoder triggers control signals from instruction bytes
- Programs avoid output values < 0x08

Result

Autonomous operation with limited output range.